

Scaled Dot-Product Attention: A Review

Abstract

Self-attention has become the dominant mechanism for sequence modeling in natural language processing, replacing recurrent architectures in most modern systems. This paper reviews the scaled dot-product attention mechanism, its computational properties, and the architectural choices that allow Transformer models to be trained in parallel across very long sequences.

1. Introduction

Recurrent neural networks process sequences one token at a time, with each hidden state depending on the previous one. This sequential dependency prevents parallelization during training and causes gradients to vanish across long sequences, since the gradient signal must pass through many multiplicative steps to reach early tokens.

Self-attention resolves both limitations by allowing every position in a sequence to attend directly to every other position in a single operation, with a path length of one between any two tokens regardless of their distance in the sequence.

2. Scaled Dot-Product Attention

Given a set of queries Q , keys K , and values V , attention is computed as the softmax of the scaled dot product between queries and keys, applied to the values. The scaling factor of one over the square root of the key dimension prevents the dot products from growing too large in magnitude, which would otherwise push the softmax function into regions with extremely small gradients.

Multi-head attention extends this mechanism by computing several attention operations in parallel, each over a smaller subspace of the representation. This allows different heads to specialize in different types of relationships between tokens, such as syntactic dependencies or coreference.

3. Positional Encoding

Because self-attention has no inherent notion of token order, position information must be injected separately. The original Transformer uses sinusoidal functions of different frequencies for this purpose, which allows the model to generalize to sequence lengths not seen during training and to learn relative-position relationships as linear transformations.

4. Conclusion

The combination of self-attention, position-wise feed-forward networks, residual connections, and layer normalization allows Transformer models to be trained efficiently at very large scale, and has become the foundation of essentially all modern large language models.